

Bytownite (DHZ Fig. 210e): crystal morphology

Habit of the bytownite drawing in Deer, Howie and Zussman (1992), Fig. 210e, on an An86 (bytownite) cell

Crystal habit reconstructed from the bytownite drawing in Deer, Howie and Zussman, *An Introduction to the Rock-Forming Minerals* (1992), Fig. 210e. The drawing shows {001}, {010}, {110} and {1-10}; the unlabelled sloping face at the bottom front (and its parallel partner behind the top of {010}) was identified by fitting as $\{-1\ 0\ 2\}$ on the 14 Å cell, which is $x = \{-1\ 0\ 1\}$ on the 7 Å morphological axes.

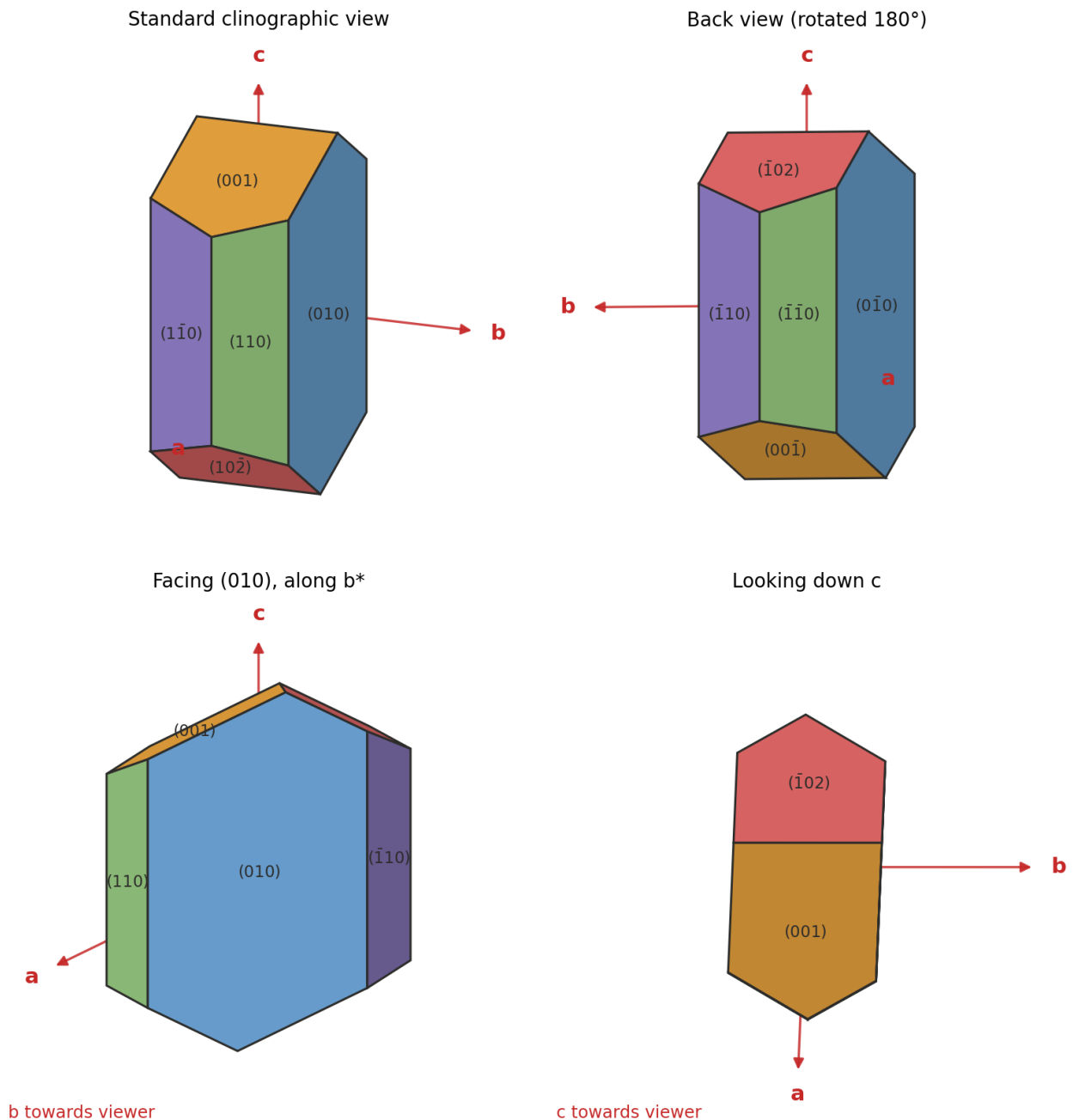


Figure 1. Orthographic views of the crystal in the frame used by SHAPE and the pygmsh classes (c along z, b in the yz plane, centre at the origin). Red arrows are the crystal axes; faces with the same colour belong to the same form.

Crystal data

Quantity	Value
SHAPE file	Bytownite_DHZ_fig210e.SHP (title BytDHZ210e, class code 2B1)
Unit cell a, b, c (Å)	8.178, 12.87, 14.19
α , β , γ (°)	93.5, 115.9, 90.63
Point group / morphology symmetry	-1 (triclinic, centrosymmetric)
Faces, corners, edges	10, 16, 24
Angle between c and the (001) normal	26.21°
pygmsh class	bytownite_dhz_fig210e (xtal_geometries_feldspars.py)

Forms

Form	Same face, 7 Å axes	Central distance	Faces	Share of surface
{0 1 0}	{0 1 0}	0.49300	2	44.6%
{0 0 1}	{0 0 1}	1.00000	2	17.2%
{1 -1 0}	{1 -1 0}	0.87000	2	14.5%
{1 1 0}	{1 1 0}	0.88700	2	12.1%
{-1 0 2}	{-1 0 1}	1.14400	2	11.8%

Central distances are those in the SHAPE file (arbitrary units). Surface shares are computed from the polyhedron built from these forms, which reproduces SHAPE's own corners to 10^{-7} .

Size and shape when the pygmsh class is called with z = 1

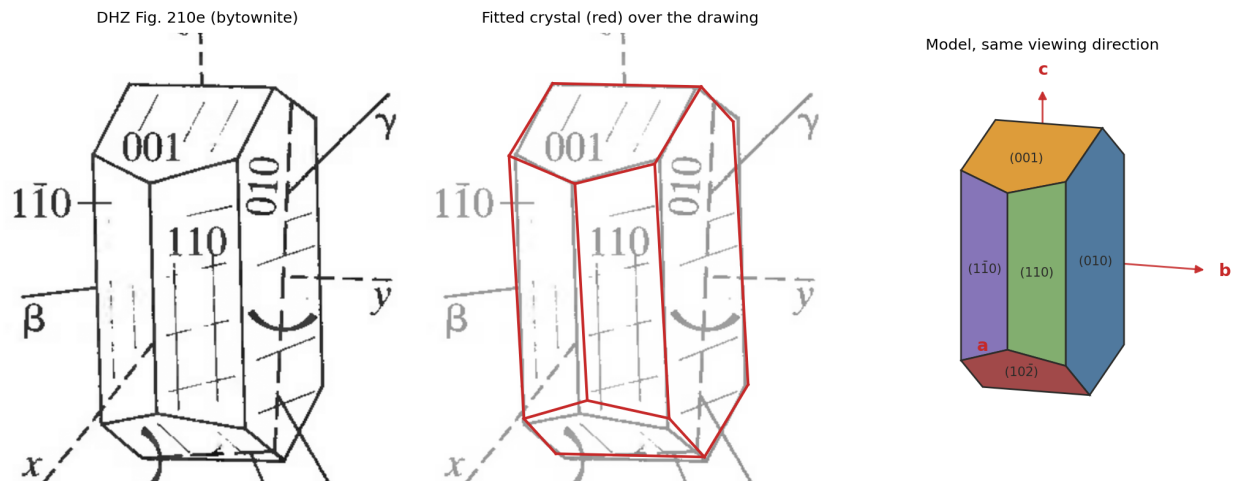
Quantity	Value
Scaling	sf = z / fsp_z with fsp_z = 1.222645 (half-length along c in SHAPE units)
Length along c	2.000 (spans -1 to +1 along z)
Width (perpendicular to c, in (010))	1.654
Thickness (perpendicular to (010))	0.806
Length : width : thickness	2.48 : 2.05 : 1
Half-extents x , y , z	0.828, 0.428, 1.000
Volume V1	1.8279
Surface area	8.8381
Equivalent-sphere radius	0.7585
Surface / volume	4.835

To match a measured length L along c use $z = L/2$; to match a measured volume V use $z = (V / V1)^{1/3}$.

Source and notes

- Cell: bytownite (Wikipedia): a, b, c = 8.178, 12.870, 14.187 Å; α , β , γ = 93.5, 115.9, 90.63°, used for An86. Face symbols are for this 14 Å cell; the 7 Å equivalents are listed in the forms table.
- Fitting: 12 corners were digitised from the drawing; the viewing direction and the face distances were fitted by least squares for several candidate faces. {-1 0 2} (= x) fitted best (15.5 px on a ~900 px drawing); the fitted viewing direction (18.1° azimuth) is close to the standard 18°26' of crystal drawings. Next best: {-2 0 3} (19.0 px) and {-1 0 1} (= y, 29.3 px).
- Distances: (001) fixed at 1.000; the others are the fitted values. Deer et al. use one schematic crystal for all compositions, so the face types are more reliable than the exact proportions.
- Scaling in the pygmsh class: fsp_z = 1.222645, the half-length along c.

Comparison with the source drawing



Left: DHZ Fig. 210e. Centre: the fitted crystal (red) drawn over it. Right: the model rendered from the fitted viewing direction.

SHAPE input file (crystal definition)

```
BytDHz210e
2B1      0      1  3.10      0
8.1780000 12.8700000 14.1870000
93.5000000 115.9000000 90.6300000
0         0         1      1.00000  0  0  0  0
0         1         0      .49300  0  0  0  0
1         1         0      .88700  0  0  0  0
1        -1         0      .87000  0  0  0  0
-1        0         2      1.14400  0  0  0  0
99        99        99      1.00000  0  0  0  0
```

Line 1: title. Line 2: crystal class code (2 = triclinic -1) and SHAPE options. Lines 3-4: a, b, c and alpha, beta, gamma. Following lines: h, k, l and central distance of each form, ending with 99 99 99. The display settings that follow in the file are not shown.